C-HI 5 Myr Observable Smoke

Date: 2026-05-10

Summary

This report records the first aligned observable smoke product for the particles-enabled high-power cluster case, `C-HI`, using the same target as the truth smoke:

- grid output: `data.0008.flt.h5`
- grid timestep: `Timestep_8`
- physical time: `4.9994 Myr`
- view: `0 deg`, projected along the simulation `y` axis
- particle output: `particles.0503.dbl`
- particle time: `4.9994 Myr`
- redshift used for this first smoke: `0.05`

The extraction was run on a `kunanyi` compute node as PBS job `690612.kunanyi-ohpc` and completed successfully. It supersedes an initial same-day trial, `690611.kunanyi-ohpc`, which used `particles.0500.dbl`; the rerun corrected this to the exactly time-aligned particle file.

What Was Tested

The supplied GRAiLE/XRAiSE bundle is present locally and contains the expected notebook, APEC table, Gaunt-factor lookup table, driver script, and notes. However, `kunanyi` currently has no Python analysis stack loaded by default: `numpy`, `h5py`, `scipy`, `astropy`, `numba`, `matplotlib`, `yaml`, and `hdf5plugin` were all absent from the baseline login environment.

For this first smoke, I therefore used a compact C extractor on the compute node rather than trying to run the full Python driver on the login node. The extractor:

- ports the supplied XRAiSE grid bremsstrahlung formula before the APEC/Gaunt correction stage
- ports the lossless `plutokore` pressure/tracer radio surface-brightness scaling for a first 150 MHz radio proxy
- reads the aligned PLUTO particle file and writes particle positions for the same 0 degree target
- writes compact 2D maps and scalar summaries for MacBook-side plotting

This is a traceable bridge smoke, not the final science observable pipeline. The full GRAiLE/XRAiSE loss-kernel path still needs a portable Python runtime on either `kunanyi` or a suitable analysis host.

Kunanyi Job

- job id: `690612.kunanyi-ohpc`
- job name: `c_hi_obs5`
- node: `n001`
- resources: `select=1:ncpus=1:mpiprocs=1`
- walltime requested: `00:30:00`
- actual extraction window: `2026-05-10T08:03:53Z` to `2026-05-10T08:04:14Z`
- extraction result: no stderr output, archive written successfully

Compact archive:

```
remote: /u/bwbrook/codex_kunanyi/c_hi_observable_smoke_2026-05-10.tar.gz
local:  build/science/kunanyi_stage_2026-05-08/c_hi_observable_smoke_690612_compact_2026-05-
sha256: 477b576e9eacb662b9ed715e3737eecbe44f7b0121f66bcc19d9adc7ec78805e
```

Scalar Smoke Metrics

Metric	Value
grid time	4.9994348189788402 Myr
particle time	4.9994348189790641 Myr
particles loaded	16056
tracer threshold	5e-3
raw 1 keV X-ray surface-brightness max	1.4382e6 mJy sr^-1
150 MHz radio proxy max	6.6204e15 mJy sr^-1
150 MHz radio luminosity proxy	5.4149e26 W Hz^-1
radio/tracer cells counted	14296426

Particle extent at the aligned snapshot:

Axis	Range
x	-8.84 to 9.01 kpc
y	-8.81 to 8.70 kpc
z	-32.24 to 32.20 kpc

Figure

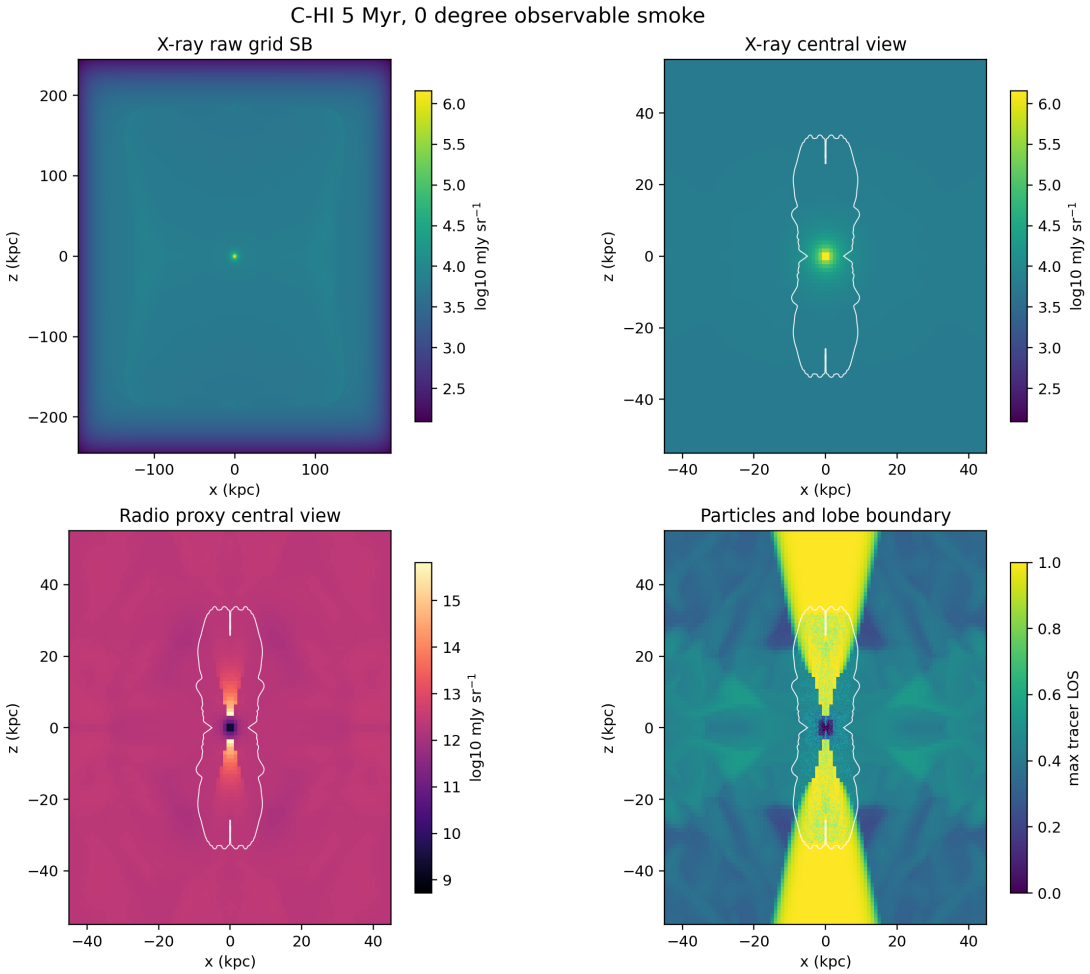


Figure 1: C-HI observable smoke maps

The white contour marks the line-of-sight maximum tracer boundary at `tracer = 5e-3`. The particle panel overlays the aligned particle positions on the same projected lobe-boundary view.

Products

The compact products are stored under `docs/science/assets/c_hi_observable_smoke_2026-05-10/`:

- `observable_summary.csv`

- particle_summary.csv
- particle_positions_0deg.tsv
- xray_1kev_raw_mjy_sr.tsv
- radio_150mhz_proxy_mjy_sr.tsv
- tracer_max_los.tsv
- x_kpc.tsv
- z_kpc.tsv
- c_hi_observable_smoke_maps.png

The extractor and plotting helper are:

- scripts/science/pluto_h5_observable_smoke_extract.c
- scripts/science/plot_observable_smoke.py
- scripts/kunanyi/c_hi_observable_smoke.pbs

Two SCP helpers were also added for repeatable MacBook-control-plane transfer through the Nectar bastion route:

- scripts/kunanyi/copy_to_kunanyi.sh
- scripts/kunanyi/fetch_from_kunanyi.sh

Interpretation

This passes the first observable smoke gate:

- the same `5 Myr`, `0 deg` grid/particle target can be processed on `kunanyi`
- PLUTO HDF5 data can be reduced into compact projected X-ray/radio products without moving multi-GB HDF5 files to the MacBook
- the particle output loads and aligns exactly with the target snapshot
- the lobe boundary is visible and consistent with the previous tracer/truth diagnostics
- the radio proxy yields a finite first luminosity-scale product

Remaining Work

The immediate next science/tooling step is to turn this bridge smoke into the proper observable pipeline:

1. build a portable Python analysis environment for the supplied XRAiSE/GRAiLE scripts, preferably on a compute-safe host
2. replace the bridge X-ray map with the full XRAiSE path, including APEC/Gaunt corrections and particle-loss kernel mapping
3. replace the lossless radio proxy with the full particle-loss-aware radio product through the `plutokore`/GRAiLE lane

4. add the first cavity-protocol stub against this same snapshot/view: ellipsoidal boundary, local ambient pressure, `pV`, `4pV`, and simple age estimates